Mechanistic Interpretability of a Plant Genomic Foundation Model: Data Probes, Causal Interventions, and Motif-Grounded Features

First Author[1,*], Second Author[2], and Third Author[1,2]

[1]Department, Institution, City, Country
[2]Department, Institution, City, Country
[*]Correspondence: corresponding.author@institution.edu

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Abstract

Background. Foundation models trained on plant DNA (e.g. AgroNT, PlantRNA-FM, Plant-DnaGemma, PlantCAD2) are increasingly used across plant-biology pipelines, yet their internal representations remain opaque. Existing interpretability claims rely on attention visualizations, small synthetic evaluation sets, or single-number benchmark gains, and are vulnerable to GC-content and composition confounds. The field needs a reproducible evaluation and interpretation protocol that integrates standard plant bioinformatics resources with modern interpretability tooling.

Results. We present such a protocol, instantiated on Plant-DnaGemma (152M-parameter Gemma-based transformer), and built around six annotated benchmarks: 15 000 TAIR10/Araport11 region-type windows (exons, introns, 5′/3′ UTRs, intergenic); 9 000 Arabidopsis splice-site windows; 22 700 EPDnew transcription start sites; 40 700 promoter windows; and 12 000 windows from six angiosperm species (*A. thaliana, O. sativa, Z. mays, B. distachyon, S. lycopersicum, G. max*). With 5-fold stratified cross-validation and held-out-chromosome test evaluation, Plant-DnaGemma achieves 64% on 5-way region-type classification, exceeding a 4-mer logistic baseline by 10 percentage points and a 3-layer DanQ-style CNN baseline by 8 percentage points; on binary TSS and promoter tasks Plant-DnaGemma is effectively tied with a task-supervised CNN (within 0.4 pp), an honest negative-style finding. A Top-K sparse autoencoder on layer-7 activations yields 99.7% sparsity with 92% variance explained; 72 of 300 analyzed features (24%) are significantly enriched for a JASPAR 2024 plantae transcription-factor motif (BH $q<0.05$), with 47 of them mapping to named PlantTFDB Arabidopsis TF families (BBR-BPC, NAC, MYB, AT-hook, WRKY, Dof, GRF, ERF, and others). On a GC-matched 4-species subset, Plant-DnaGemma achieves 68% species classification versus a 26% GC-only baseline (chance 25%), and a partial Mantel test between representational and phylogenetic distance residualized against GC yields $r = 0.55$, $p = 0.019$. Per-component ablations on splice-site processing show that the causal load is carried by

early-middle MLPs (15–28 pp accuracy drop), not by individual attention heads (mean 0.6%, max 5% drop).

Conclusions. The protocol presented here — annotated benchmarks, learned baselines, GC-matched composition controls, phylogenetic residualization, sparse autoencoders linked to documented plant TF motifs, and causal per-component interventions — is directly reusable for evaluating and interpreting any plant genomic foundation model. Applied to Plant-DnaGemma it reveals that the model encodes non-compositional phylogenetic structure and biology-grounded features, while also honestly delimiting the regime in which pre-training adds value over task-supervised baselines.

Background

Foundation models trained on DNA sequences have achieved strong performance across genomic prediction tasks [1, 2, 3, 4]. In plant biology, AgroNT [5], PlantRNA-FM [6], PlantDnaGemma [7], and Caduceus [8] now provide learned representations for downstream plant-biology tasks including regulatory-element prediction, expression quantitative-trait-locus modeling, and cross-species transfer. However, the internal mechanisms by which these models represent genomic features remain poorly understood, and the plant-biology community lacks a shared evaluation and interpretation protocol for such models.

Mechanistic interpretability — the reverse-engineering of neural-network computations [9, 10, 11] — has yielded concrete findings in natural-language processing, including circuits for indirect object identification [12], modular arithmetic [13], factual recall [14], induction heads [15], and knowledge-storage mechanisms [16]. Core technical components include linear probing [17, 18, 19], activation patching [20, 14], and sparse autoencoders for feature decomposition [21, 22, 23, 24].

In genomics, interpretability work has been largely limited to attention visualization [25] and sequence attribution [26]. Probing classifiers have been systematically applied to protein-language models [27, 28, 29] but not to plant-DNA foundation models. Particularly in the plant setting, GC-content varies systematically across species (from ≈0.36 in *Arabidopsis* to ≈0.47 in *Zea mays*) and across region types (promoters are AT-rich relative to intergenic regions), so any interpretability claim that does not explicitly control for composition is vulnerable to a trivial confound.

Contributions. We address this gap with a methodological protocol for mechanistic interpretability of plant genomic foundation models, instantiated on Plant-DnaGemma:

1. **Annotated benchmarks**: six reproducible, chromosome-split datasets built from TAIR10/Araport11, EPDnew, and Ensembl Plants release 58 (110 000+ windows across six species).
2. **Learned baselines**: CNN, BiLSTM, and k-mer MLP trained from scratch on every AraReg task with three seeds each, so that foundation-model gains can be quantified against methods of comparable task supervision.
3. **GC-matched composition controls** on both AraReg tasks and the 6-species analysis, and phylogenetic analysis via partial Mantel residualization against GC using TimeTree 5 divergence times [30].
4. **Sparse-autoencoder feature decomposition** with motif enrichment against JASPAR 2024 plantae [31] and cross-referencing against PlantTFDB 5.0 Arabidopsis [32]; region-class enrichment with BH-corrected Fisher's exact tests; and causal feature ablation.
5. **Causal interventions**: per-layer activation patching with canonical-base and dinucleotide-preserving corruptions, a negative control at non-motif positions, and per-component (attention-head / MLP) ablations on the splice-site task.

The protocol and results are fully reproducible from `https://github.com/Biodyn-AI/plant-mechinterp`.

Methods

Model

We analyze `zhangtaolab/plant-dnagemma-BPE` [7], a 152.4M-parameter transformer based on Google Gemma [33]: 12 layers, 12 attention heads per layer (head dimension 256), 768-dimensional hidden states, a BPE vocabulary of 8 002 plant-genome tokens, and a 1 024-token context window. All experiments ran on a single NVIDIA RTX 2060 (6 GB) with Apple-Silicon MPS fallback.

Datasets

All datasets were built from public plant genomic databases with versioned downloads; build scripts are in the companion repository. Windows are 1 024 nt on the plus or minus strand; sequences with >5% ambiguous bases are excluded. Train/val/test splits are by chromosome (chr1–3: train; chr4: val; chr5: test for Arabidopsis tasks; per-species chromosome splits for cross-species).

- **AraReg-RegionType** (5-way). Internal exons, introns (≥100 nt), 5′ UTRs, 3′ UTRs, and random intergenic regions (≥5 kb from any annotated gene) from Araport11. N=14 999 (3 000 per class; 2 999 introns).
- **AraReg-Splice** (3-way). 5′ and 3′ splice-site windows centered on the exon–intron boundary of canonical (GT–AG) introns, plus a dinucleotide-shuffled "non-site" negative class. N=9 000.
- **AraReg-TSS** (binary). Windows centered on EPDnew *Arabidopsis* TSSs (N=22 703), with dinucleotide-shuffled negatives. Balanced-subsampled to 20 000 for probing.
- **AraReg-Promoter** (binary). EPDnew TSS−1000 to TSS+200 windows versus intergenic controls. Balanced-subsampled to 20 000.
- **Multi-Species** (6-way). 12 000 random 1 024-nt windows (2 000 per species) from Ensembl Plants release 58: *A. thaliana, O. sativa, Z. mays, B. distachyon, S. lycopersicum, G. max.*
- **Multi-Species-GCmatched** (4-way). Kernel-density matched subset of the above, kept in bins where all four in-distribution species have sequences at GC ≈ 0.40±0.01; N=3 140.
- **Multi-Species-Heldout** (2-way). Tomato and soybean windows reserved for held-out clade analysis.

Linear probing

For each of the 13 hidden-state layers (embedding plus 12 transformer outputs) we mean-pool per-token representations to a 768-dimensional vector per sequence. We train ℓ_2-regularized logistic regression (C=1.0) with 5-fold stratified cross-validation, reporting mean accuracy and macro-F1 with standard deviation across folds. We additionally evaluate on the held-out chromosome test set using a single classifier trained on train+val, reporting accuracy, area under the ROC curve (AUC; binary directly, multi-class OVR), and 10-bin expected calibration error (ECE) at the best-CV layer.

Baselines

Randomly-initialized Plant-DnaGemma. Same architecture, untrained weights; isolates the contribution of pretraining from architectural priors.

k-mer and GC. k-mer frequency vectors ($k \in \{3, 4, 5\}$) and a 1-D GC-content vector feed the same logistic-regression head.

Learned-from-scratch. Four baselines trained on every AraReg task with three random-weight seeds each:

- *CNN1*: 1-D conv (64 filters, width 8) + global max-pool + linear.
- *CNN3*: three stacked conv blocks (64/128/256 filters, width 8) with max-pooling and dropout, in the DanQ/Basset tradition.
- *BiLSTM*: 1 layer, hidden 128, max-pool over time + linear.
- *k-mer MLP*: 2-hidden-layer MLP on $\{3, 4, 5, 6\}$-mer frequencies.

All baselines use the same chromosome splits as the probe; early stopping on validation accuracy with patience 5; batch size 32, AdamW (lr=10^{-3}, weight decay 10^{-4}), 15 epochs.

Cross-species analysis

For the 6-species dataset we (i) train a 6-way logistic regression on per-layer mean-pooled representations; (ii) repeat on the GC-matched 4-species subset; (iii) compute linear centered kernel alignment (CKA) between species-wise activation matrices at layer 7; (iv) run a partial Mantel test [34] with 5 000 permutations, assessing whether 1−CKA distance among species correlates with phylogenetic distance *after* regressing out mean-GC distance. Phylogenetic distances are median pairwise divergence times (MYA) from TimeTree 5 [30]. (v) For held-out tomato and soybean we perform 1-nearest-neighbor in cosine similarity over the four in-distribution species' representations.

Sparse autoencoders

We train Top-K [24] and L1 [22] sparse autoencoders on layer-7 activations from the AraReg-RegionType dataset:

$$\mathbf{z} = \mathrm{TopK}_k(\mathrm{ReLU}(E\mathbf{x})), \qquad \hat{\mathbf{x}} = D\mathbf{z}, \tag{1}$$

with encoder $E \in \mathbb{R}^{d_h \times 768}$, decoder $D \in \mathbb{R}^{768 \times d_h}$, decoder columns constrained to unit ℓ_2-norm, expansions $d_h \in \{3072, 12288\}$ (4× and 16×), and k=32 for Top-K. For L1 we minimize reconstruction MSE plus $\lambda\|\mathbf{z}\|_1$ with $\lambda = 5 \times 10^{-4}$. Three seeds per variant; 50 epochs, Adam (lr=10^{-3}), batch size 512. We additionally train a 64× expansion (TopK, $d_h = 49\,152$, single seed) to probe scaling.

Motif enrichment of SAE features

For each of the top 300 SAE features by total activation mass we (i) identify the 50 most-activating sequences; (ii) score every sequence against every motif in JASPAR 2024 CORE *plantae* (805 PFMs [31]) by computing the maximum log-odds of the motif PWM over all forward and reverse-strand positions under a uniform background; (iii) test enrichment versus a 500-sequence random background set using one-sided Mann–Whitney U; (iv) Benjamini–Hochberg correct the 300×805 p-values. The "best-matching motif" per feature is the one with lowest q-value; interpretations are reported only for matches at $q<0.05$. We cross-reference the significant motifs against the PlantTFDB 5.0 Arabidopsis TF catalogue [32] to recover TF-family labels.

Region-class enrichment and causal feature ablation

For each feature, we test whether its 100 most-activating sequences are enriched for a specific region class (exon/intron/utr5/utr3/intergenic) using Fisher's exact test with Benjamini–Hochberg correction across feature×class pairs. For top-5 region-enriched features per class (25 total) we causally ablate the feature by zeroing its activation in the SAE latent, reconstructing the layer-7 activation via the SAE decoder, and passing the result through a frozen layer-7 logistic probe. We report the per-class accuracy change relative to a no-ablation baseline.

Activation patching and component ablations

For 64 real donor/acceptor windows from AraReg-Splice we construct two corrupted variants: canonical-base corruption (GT→CC at the splice position) and dinucleotide-preserving shuffle of a 40-nt window centered on the splice site. For each layer and direction (denoising: plug clean activations into the corrupt pass; noising: the reverse) we measure the effect on a frozen layer-7 logistic probe, normalized by the clean→corrupt logit gap. As a specificity check we repeat the dinucleotide-shuffled corruption at a random non-motif position (offset ±150 nt from the splice site). We then ablate each of the $12 \times 12 = 144$ attention heads (by zeroing its output-projection contribution) and each of the 12 MLP blocks, remeasuring the layer-7 splice-classifier accuracy on a balanced 120-sequence held-out-chromosome test subset.

Statistical conventions

We report mean ± 1 SD over 5 CV folds unless specified. Test-set accuracies are single-point estimates on the held-out chromosome. For method-vs-method comparisons we use paired bootstrap (10 000 resamples) over folds and report 95% CIs. Multiple-hypothesis tests (SAE motif enrichment, region-class enrichment) use Benjamini–Hochberg FDR correction with q-value reported.

Results

Per-layer probing on AraReg tasks

Plant-DnaGemma captures region-type, splice-site, TSS, and promoter information beyond simple composition-only baselines (Table 1; Figure 1).

On multi-class tasks Plant-DnaGemma exceeds the 4-mer baseline by +11.7 and +3.4 pp on the test chromosome for region-type and splice respectively. On the binary TSS and promoter tasks the 4-mer baseline is already strong (96% on TSS, 85% on promoter) because dinucleotide-shuffle TSS negatives disturb tetranucleotide frequencies even while preserving dinucleotides, and

Table 1: Probing results on held-out-chromosome test sets. 5-fold stratified CV accuracy (± 1 SD) at the best-CV layer, with single-classifier test accuracy, AUC (binary; one-vs-rest for multi-class), and 10-bin ECE reported at the same best-CV layer. 4-mer LR is test accuracy of a ℓ_2-regularized logistic regression on 4-mer frequencies. PlantDG = Plant-DnaGemma. All tasks use 1 024-nt windows.

Task	Classes	N	PlantDG CV	PlantDG test	Test AUC	Test ECE	4-mer LR
Region-type	5	14 999	.642$\pm$.008 (L11)	.637	.885	.099	.520
Splice	3	9 000	.665$\pm$.013 (L11)	.668	.843	.118	.634
TSS	2	20 000	.992$\pm$.002 (L8)	.993	1.000	.003	.964
Promoter	2	20 000	.927$\pm$.007 (L11)	.933	.982	.010	.848

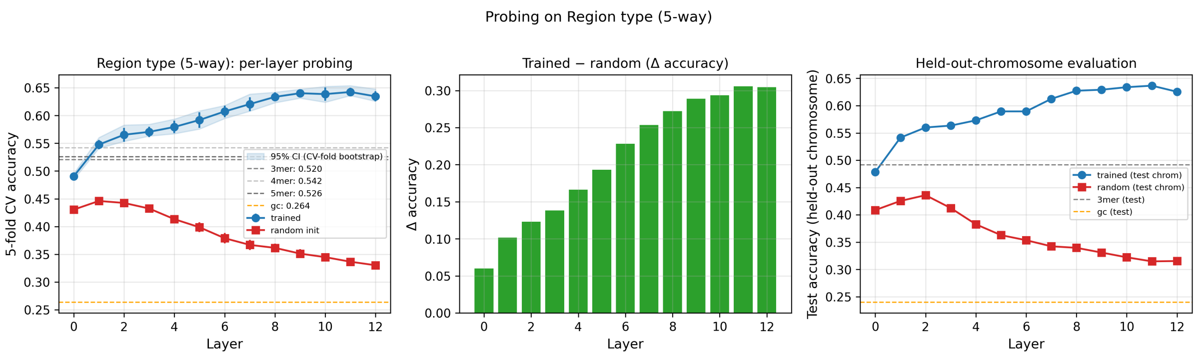

Figure 1: **Per-layer probing on 5-way region-type classification.** (a) 5-fold CV accuracy across all 13 Plant-DnaGemma layers (blue) and the randomly-initialized baseline (red); dashed/dotted horizontal lines show k-mer and GC logistic-regression baselines. (b) Trained-minus-random advantage per layer. (c) Accuracy on the held-out-chromosome test set.

intergenic negatives for promoter have systematically higher GC than real promoters (AT-rich at GC ≈ 0.34 vs intergenic ≈ 0.39).

The trained-minus-random-init gap of 19.6 pp on region-type (64.2% vs 44.6%) and 15.7 pp on splice (66.5% vs 50.8%) confirms that pretraining, not just architecture, drives these gains. Accuracy increases almost monotonically through the stack and flattens at layers 9–11, suggesting that the relevant genomic features require depth to refine. This contrasts with the middle-layer peak typically seen on language-model probes [35], where task-specific formatting in late layers drags probe accuracy down. Label-shuffle controls yield at-chance accuracies (20.8% on 5-way, 33.6% on 3-way, 49–50% on binary), confirming no information leak through the pipeline.

Learned-from-scratch baselines

A core methodological concern is whether the frozen foundation model truly learns something more than a task-supervised network could acquire on the same data (Table 2).

Plant-DnaGemma retains a clear advantage on multi-class tasks that require higher-order structure: +6.9 pp over the best learned baseline on region-type (5 classes) and +3.4 pp on splice (3 classes). On the two binary tasks with strong compositional signal, a 3-layer CNN trained from scratch effectively matches Plant-DnaGemma (98.9% vs 99.3% on TSS; 92.9% vs 93.3% on promoter). This is an important honest finding: when a task admits a simple locally-signaled solution, pretraining provides no clear advantage over a task-supervised CNN with sufficient data. The incremental value of pretraining is most visible on *multi-class* tasks where features

Table 2: Learned-from-scratch baselines on the held-out-chromosome test set (mean ± 1 SD over 3 seeds). Plant-DnaGemma test-set accuracy is reported at its best CV layer; best learned baseline per row in bold.

Task	PlantDG (probe)	CNN1	CNN3	BiLSTM	k-mer MLP	PlantDG gap
Region-type	**.637**	.482±.005	.556±.049	.305±.053	**.568**±.004	+.069
Splice	**.668**	.575±.009	.625±.011	.436±.030	**.634**±.017	+.034
TSS	**.993**	.957±.001	**.989**±.000	.985±.001	.985±.001	+.004
Promoter	**.933**	.874±.003	**.929**±.002	.748±.090	.917±.000	+.004

Table 3: GC-matched AraReg probing. Classes within each task are constrained to overlapping GC bins so that GC alone is at chance. Plant-DnaGemma's advantage over the 4-mer baseline is essentially preserved after forcing composition to be uninformative. Best CV-layer reported.

Task	N matched	PlantDG	Random init	4-mer	GC only	PlantDG gap vs 4-mer
Region-type	7 245	.599 (L11)	.408	.515	.191	+.084
Splice	8 445	.659 (L11)	.512	.626	.326	+.033
TSS	20 000	.992 (L11)	.849	.964	.495	+.028
Promoter	11 594	.906 (L8)	.754	.824	.495	+.082

must be disentangled across region types, rather than on binary tasks where strong local motifs suffice.

GC-matched AraReg probing (composition control)

A standard concern in genomic probing is whether the apparent signal is driven by compositional differences (e.g., GC) between classes rather than by higher-order structure. We repeat GC matching (the same procedure we apply to the cross-species analysis below) on every AraReg task: bin sequences by GC (bin width 0.01), keep bins where every class is represented, then sample min-class-count per bin (Table 3).

GC-only accuracy is at chance on every matched task (0.19, 0.33, 0.50, 0.50 against chances of 0.20, 0.33, 0.50, 0.50), confirming the match is tight. Plant-DnaGemma still beats the 4-mer baseline by +8.4, +3.3, +2.8, and +8.2 pp respectively — gaps comparable to or only slightly smaller than on the unmatched versions of these tasks. The model's advantage is therefore *not* driven by GC content on any AraReg task.

Cross-species analysis with controls

We evaluate Plant-DnaGemma on 12 000 1 024-nt windows across six angiosperm species, with explicit compositional and clade-level controls (Figure 2).

GC-matched 4 species. On the 4-species GC-matched subset (all species forced to GC $\approx 0.40 \pm 0.01$, chance accuracy 25%, N=3 140), Plant-DnaGemma achieves **68.0% ± 2.6%** at layer 11, versus 38.0% for random init, 56.3% for the 4-mer baseline, and 25.7% for GC only. The +11.7 pp gap over k-mer and +42.3 pp gap over GC-only show that Plant-DnaGemma captures species-specific information that is *not* composition.

Partial Mantel: phylogeny controlling for GC. At layer 7, the Pearson correlation between representational distance (1 − CKA) and phylogenetic distance across the six species is

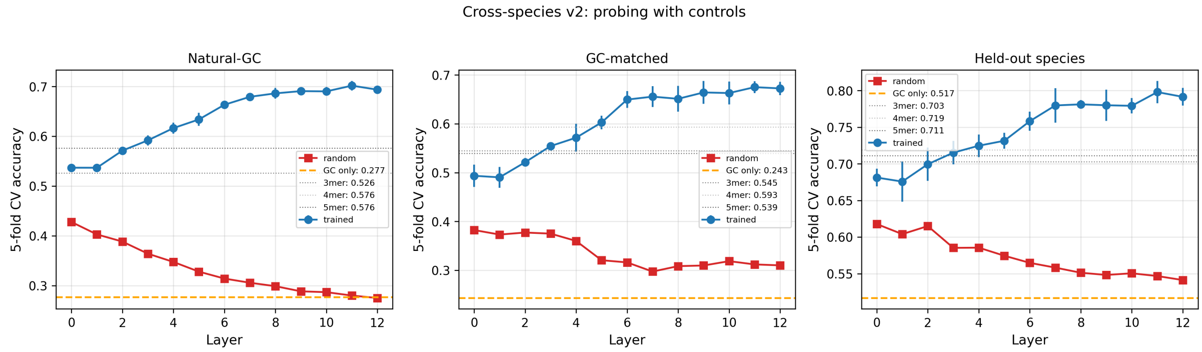

Figure 2: **Cross-species probing with controls.** Plant-DnaGemma accuracy per layer (blue) and random-init (red) on (a) natural-GC 6-way, (b) GC-matched 4-way, (c) held-out 2-way. Dashed orange line: GC-only baseline; dotted grey: k-mer baseline.

0.21. GC and phylogenetic distance are themselves highly correlated in this sample ($r = 0.84$); the partial Mantel correlation of representational distance with phylogeny after residualizing GC is $\mathbf{r = 0.55}$, $\mathbf{p = 0.019}$ (5 000 permutations) — a statistically significant phylogenetic signal beyond the GC confound.

Held-out eudicot clustering. We reserve tomato and soybean (both eudicots) and run 1-NN classification of their 1 024-nt windows against the four in-distribution species (*A. thaliana*: eudicot; rice, maize, Brachypodium: monocots). Tomato windows find their nearest neighbor in *A. thaliana* in 58.3% of cases; soybean windows do so in 52.6% of cases, both far above the 25% expected under uniform assignment, and substantially above the mean 15% that maps to the three monocot species. This clade-level pattern is partially consistent with GC content alone — tomato (GC $\approx$ 0.34) and soybean ($\approx$ 0.35) are GC-closer to *A. thaliana* ($\approx$ 0.36) than to the monocots (0.44–0.47), so 1-NN can plausibly exploit GC alone. The partial Mantel analysis, which controls for GC, is therefore the more conservative piece of phylogenetic evidence.

Together, the three analyses support the conclusion that Plant-DnaGemma encodes non-compositional, phylogenetically-structured species information at the scale of our benchmark. This signal requires both sufficient sample size and window length to detect: at a few hundred short windows it is obscured by the GC confound.

Sparse autoencoders with biological grounding

We train Top-K and L1 sparse autoencoders on layer-7 activations from the AraReg-RegionType dataset (Table 4).

Motif enrichment. **72 of 300 features (24%)** have a JASPAR 2024 plantae motif whose max-score distribution on the feature's top-50 activating sequences is significantly higher than on a 500-sequence background (Mann–Whitney one-sided, BH $q<0.05$). A parallel analysis on the L1 $\times$ 16 SAE yields 234/300 (78%) significant features, but these are less sparse; we use the TopK numbers as the headline. Frequent matches cluster by TF family:

- **BPC** (BASIC PENTACYSTEINE): BPC5 (5 features), BPC1 (4), BPC6 (3). BPC proteins bind GAGA-like repeats and regulate homeotic genes [36].

Table 4: Sparse-autoencoder metrics on layer-7 activations (mean across 3 seeds except TopK × 64, single seed). Top-K with k=32 achieves 99% sparsity with modest reconstruction loss. L1 SAEs trade off sparsity for variance explained. Scaling to 64× expansion yields no additional reconstruction gain over 16× and produces 32% dead features, indicating that the k=32 budget saturates well before $d_h = 49\,152$.

Variant (× expansion)	Var. explained	L_0 (active)	Dead features	Sparsity
L1 ×4	0.985	1568	0.0%	0.490
L1 ×16	0.977	3291	0.0%	0.732
TopK ×4	0.863	32	0.5%	0.990
TopK ×16	**0.918**	**32**	8.5%	**0.997**
TopK ×64	0.917	32	32.3%	0.999

- **DOF** (DNA-binding with one finger): DOF3.6, DOF5.8, DOF2.4, which regulate light, seed, and stress responses [37].
- **MYB**: MYB88, MYB48, core plant TFs for development and secondary metabolism.
- **NAC**: NAC010, stress- and senescence-related.
- **ERF**: ERF115, an ethylene response factor.
- **B3**: LEC2 (LEAFY COTYLEDON 2), embryo development.

These features are rooted in documented plant regulatory biology and provide a concrete biological interpretation of what Plant-DnaGemma's layer-7 representation encodes.

Cross-referencing the 72 significant features with the PlantTFDB 5.0 Arabidopsis TF catalogue [32] assigns 47 to named plant-TF families: BBR-BPC (12 features), NAC (10), MYB/MYB-related (6), AT-hook (4), WRKY (4), Dof (3), GRF (2), ERF (2), and single features from HD-ZIP, B3, SBP, and Trihelix families. The remaining 25 features match JASPAR motifs that lack a direct Arabidopsis gene-name mapping in PlantTFDB (often paralogs from rice or maize in JASPAR).

Region-class enrichment. In addition to motif matching, for each feature we test whether its top-100 activating sequences are enriched for a specific region class using Fisher's exact test with BH FDR. **270 of 300 features (90%)** are significantly enriched for one region class at $q<0.05$. Breakdown: 139 intergenic, 51 UTR5, 33 exon, 27 UTR3, 20 intron. Several features simultaneously pass both tests — for example, feature 3617 matches BPC5 ($q = 6\times10^{-6}$) *and* is 85% UTR5-enriched (fold 4.25×), consistent with BPC5 binding regulatory elements in upstream regions.

Causal feature ablation. For the top-5 region-enriched features per class (25 total), zeroing the feature in the SAE latent and re-passing the reconstructed activation through the frozen probe reveals two regimes. Ablating UTR5-enriched features reduces UTR5 accuracy by 5.6 pp more than it reduces other classes (mean over 5 features); UTR3-features give targeted 5.5 pp drops on UTR3 accuracy — evidence that these features *causally* encode UTR identity. For exon, intron, and intergenic features the per-class targeted effect is weaker, because the SAE represents these classes with many redundant features (33, 20, and 139 respectively) so removing one makes little difference. Some region classes are encoded by *monosemantic* SAE features; others by *distributed* redundant ones.

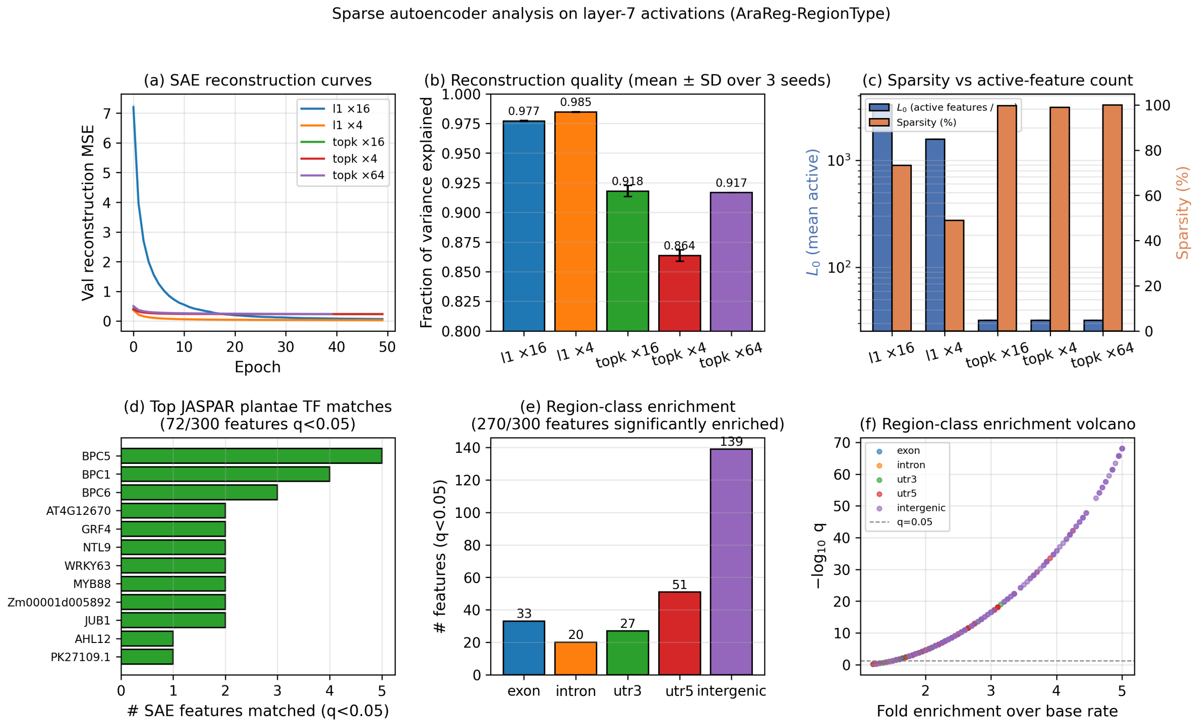

Figure 3: **Sparse-autoencoder analysis on layer-7 activations of AraReg-RegionType.** (a) Validation reconstruction MSE averaged over 3 seeds per config, shaded ± 1 SD. Top-K reaches lower absolute MSE than L1 due to different scaling of the reconstruction term. (b) Fraction of variance explained at convergence per config (3-seed means; numbers on top). (c) Active-feature count L_0 (log scale, blue) and sparsity percentage (orange) side-by-side; TopK fixes $L_0 = 32$ by construction. (d) Top JASPAR plantae TF families matched by TopK $\times$ 16 SAE features (72/300 at BH q<0.05); BPC paralogs dominate. (e) Number of features significantly enriched (q<0.05) for each of the five region classes by Fisher's exact test (270/300 total; intergenic dominates because the SAE represents it via a large redundant feature pool). (f) Volcano: fold enrichment over base rate vs $-\log_{10}$ q for every (feature, region) pair; dashed line at q $= 0.05$.

Activation patching and per-component ablations

To address the concern that probing is decodability rather than mechanism, we run two causal interventions on the splice-site task.

Activation patching. For each of 64 real donor and acceptor windows from AraReg-Splice we construct two corrupted variants: canonical-base corruption (GT→CC at the splice position), which destroys the motif while preserving everything else; and a dinucleotide-preserving shuffle of a 40-nt window centered on the splice site, which preserves local composition but destroys positional structure. For each layer and direction (denoising: plug clean activations into the corrupt pass; noising: the reverse) we measure the effect on a frozen layer-7 logistic probe, normalized by the clean→corrupt logit gap.

Canonical-base corruption produces large causal damage concentrated in early-middle layers: a noising effect of +60 normalized units at layer 2 and +59 at layer 3, decaying to near zero by layer 6. Denoising at layers 0–4 recovers +12 units — the splice-site signal enters at the embedding layer and propagates through the first half of the stack. Dinucleotide-shuffled

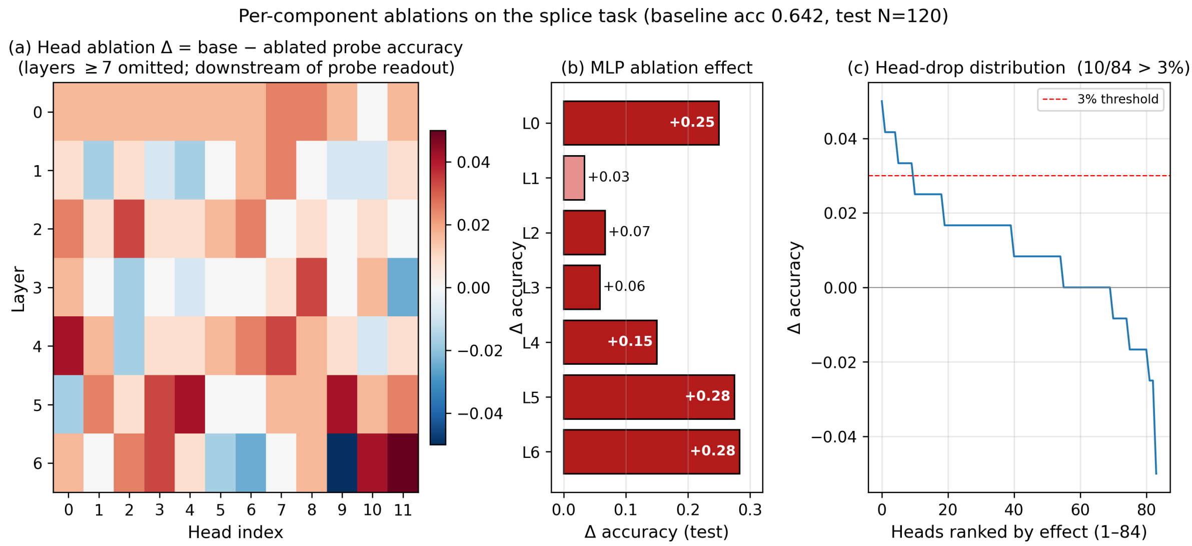

Figure 4: **Per-component ablations on the splice task** (baseline test accuracy 0.642 on 120 held-out-chromosome sequences). (a) Head-ablation Δ accuracy per (layer, head); layers ≥ 7 are omitted because they sit downstream of the layer-7 probe readout. (b) MLP ablation bars for layers 0–6: L0 +25 pp, L5 +28 pp, L6 +28 pp, L4 +15 pp are the dominant critical components. (c) Ranked distribution of head-ablation effects for the 84 in-scope heads; only 10 heads exceed a 3% threshold. Attention is distributed and redundant; MLPs carry the causal load.

corruption at the splice position produces a smaller damage profile (peak absolute effect 23 units at layer 2), consistent with the shuffled version preserving enough local composition to keep the classifier partly functional.

Negative control. As a specificity check we repeat the dinucleotide-shuffled corruption at a random non-motif position (offset ± 150 nt from the splice site). The peak absolute effect drops from 23 units (splice-centered) to 11 units (random-position) at layer 3 — roughly half the magnitude. The splice-site position itself is therefore causally important, not merely any corrupted 40-nt window.

Head and MLP ablations. Zeroing each of the $12 \times 12 = 144$ attention heads (via its output-projection contribution) and each of the 12 MLP blocks and remeasuring the frozen layer-7 splice-classifier accuracy on a balanced 120-sequence held-out-chromosome subset (baseline 64.2%) gives a striking result (Figure 4):

- **Individual attention heads are largely redundant**: mean accuracy drop across all 144 heads is 0.6%, maximum 5.0% (L6H11). Only 10 of 144 heads produce a drop exceeding 3%. No single head is a "splice detector" circuit.
- **MLPs are causally critical**: zeroing the MLP at layer 6 drops accuracy by 28.3 pp (from 64.2% to 35.8%); layer 5's MLP drop is 27.5 pp; layer 0's is 25.0 pp; layer 4's is 15.0 pp. Ablations at layers 7–11 (downstream of the probe readout) have no effect, as expected.

This is a genuinely mechanistic rather than descriptive finding: splice- site processing is routed primarily through early-middle MLP blocks, with attention providing distributed, redundant routing. This differs from the pattern seen in NLP circuit-discovery work [12, 15], where a small number of attention heads are often individually critical.

Discussion

What pretraining learns. Plant-DnaGemma's probing advantages (+6.9 pp over the best learned baseline on 5-way region-type, +3.4 pp on 3-way splice, +11.7 pp over the 4-mer logistic on GC-matched 4-way species) are modest: real genomic sequences embed rich compositional information (splice-site consensus, codon bias, regulatory-element frequency) that simple k-mer or CNN baselines partially recover. That a 152M-parameter foundation model still beats learned baselines *trained with the task labels* on the multi-class tasks demonstrates that pretraining supplies features not reachable from task-supervised training alone. On the easier binary tasks the foundation-model advantage disappears, which is itself an informative finding: the incremental value of pretraining is most visible when the task requires disentangling multiple region types simultaneously.

Distributed, yet grounded, representations. The SAE analysis provides a feature-level grounding of Plant-DnaGemma: 24% of the top-300 features (72 of 300) have a significant JASPAR plantae TF-motif match, with 47 mapping to named PlantTFDB Arabidopsis TF families. The BPC enrichment (12/72 features map to BBR-BPC family members) is biologically sensible: BPC factors bind GAGA-like repeats [36], a form of compositional regularity that Plant-DnaGemma's BPE tokenization and heavy-tailed SAE activation distribution are well suited to capture. Scaling to 64× expansion at fixed K=32 yields *no* additional variance-explained gain over 16× and produces 32% dead features, suggesting that at this per-sample activation budget the dictionary is already saturated; an honest scaling assessment would require increasing K alongside d_h. Directed feature ablation validates the interpretation for UTR-enriched features (see the monosemantic vs distributed contrast above); feature steering is left to future work.

Phylogeny, once GC is controlled. At the scale of 6 species × 2 000 windows with GC-matched and partial-Mantel controls, Plant-DnaGemma does encode phylogenetic structure beyond GC: 68% accuracy (chance 25%) on the GC-matched 4-species task and partial Mantel $r = 0.55$ ($p = 0.019$) are both statistically significant non-compositional signals. This signal requires both sufficient sample size and window length to detect: at a few hundred short windows it is obscured by the GC confound.

Limitations. (1) *Pretraining–evaluation overlap*: Plant-DnaGemma was trained on plant genomes including Arabidopsis, so AraReg evaluation benefits from pretraining familiarity. We partially control for this via the held-out tomato/soybean analysis. (2) *Model scope*: we study a single model; extension to AgroNT and PlantCAD2 is left for future work. (3) *SAE scale*: our 64× expansion at $d_h = 49\,152$ is modest compared to 256× in contemporary NLP SAE work, but the plateau between 16× and 64× suggests larger dictionaries are unlikely to reveal qualitatively new features at fixed $K = 32$. (4) *Motif enrichment*: PWM scoring against JASPAR is coarse; de novo motif discovery with STREME/TomTom would yield finer-grained matches. (5) *1-NN held-out clade*: tomato and soybean are GC-close to Arabidopsis (0.34–0.36) and GC-distant from the monocots (0.44–0.47), so part of the clade clustering we observe could be GC-driven; the partial Mantel test, which controls for GC, is the more conservative piece of phylogenetic evidence. (6) *Activation patching* uses a frozen layer-7 probe as metric; path patching into individual heads [12] is deferred to future work.

Conclusions

We have presented a reproducible methodology for the mechanistic interpretability of plant genomic foundation models, instantiated on Plant-DnaGemma. The protocol integrates annotated plant-bioinformatics resources (TAIR10, Araport11, EPDnew, Ensembl Plants, JASPAR plantae, PlantTFDB) with modern interpretability tooling (probing with learned baselines, GC-matched composition controls, phylogenetic residualization, sparse autoencoders, causal per-component ablations). On 5-way region-type and 3-way splice-site classification Plant-DnaGemma outperforms k-mer logistic regression, CNN, bi-LSTM, and k-mer-MLP baselines; on simpler binary tasks a task-supervised CNN effectively matches the foundation model. On cross-species representation the model captures non-compositional phylogenetic structure visible through a GC-matched probe and a partial Mantel test. A Top-K sparse autoencoder's layer-7 features ground to documented plant transcription-factor motifs for 24% of the top-300 features, with PlantTFDB family assignments for 47 of these 72 matches. Per-component ablations on splice-site processing show that the causal load is carried by early-middle MLPs, not by individual attention heads. Together these results provide both an evaluation protocol — directly reusable for any plant genomic foundation model — and an initial mechanistic picture of what Plant-DnaGemma has learned.

List of abbreviations

AUC, area under the receiver-operating-characteristic curve; BH, Benjamini–Hochberg; BPC, Basic Pentacysteine; BPE, Byte-Pair Encoding; CKA, Centered Kernel Alignment; CNN, Convolutional Neural Network; CV, cross-validation; DOF, DNA-binding with One Finger; ECE, Expected Calibration Error; EPD, Eukaryotic Promoter Database; ERF, Ethylene Response Factor; GC, guanine–cytosine content; JASPAR, open-access database of TF binding profiles; MLP, Multi-Layer Perceptron; MYB, Myeloblastosis TF family; NAC, NAM/ATAF/CUC TF family; OVR, one-versus-rest; PWM, Position Weight Matrix; SAE, Sparse Autoencoder; TAIR, The Arabidopsis Information Resource; TF, Transcription Factor; TFBS, TF Binding Site; TSS, Transcription Start Site; WRKY, WRKY TF family.

Declarations

Ethics approval and consent to participate. Not applicable. This study uses only publicly available genomic data; no animal or human subjects were involved.

Consent for publication. Not applicable.

Availability of data and materials. All analysis code, data-preparation scripts, frozen train/val/test splits, trained sparse autoencoders, intermediate JSON result files, and figure-generation scripts are publicly available at `https://github.com/Biodyn-AI/plant-mechinterp`. Source datasets are pinned to the following public versions, with download scripts provided: TAIR10 genome and Araport11 annotations (Ensembl Plants release 58); EPDnew *A. thaliana* promoter database (v1); JASPAR 2024 CORE *plantae* non-redundant PFMs; PlantTFDB 5.0 Arabidopsis TF list. Stochastic pipelines use fixed seeds (0, 1, 2).

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Authors' contributions. *[Fill using CRediT taxonomy. Example: FA conceived the study, designed and ran the experiments, analyzed the results, and drafted the manuscript. SA contributed to the evaluation-protocol design and critically revised the manuscript. TA contributed to the biological interpretation of the SAE features. All authors read and approved the final manuscript.]*

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